

Exercises

Exercise 4: Cleaning, summarising and linking data

Read Chapter 3 to help you complete the questions in this exercise.

This exercise picks up exactly where Exercise 3 left off and uses the same `cardiac` dataframe. If you are starting a fresh R session, either continue with the R script you wrote for Exercise 3 or run the two lines below before you go any further. They are the import from Exercise 3, Q5 and the `Fsex` factor from Exercise 3, Q7.

You will notice that `smoking` is still sitting there as an integer. That is deliberate, and there is an optional extension at the end of this exercise that deals with it rather more elegantly than typing labels out by hand.

```
cardiac <- read.table('data/cardiacdata.txt', header = TRUE, sep = "\t",
                     na.strings = "NA", stringsAsFactors = TRUE)

cardiac$Fsex <- factor(cardiac$sex, levels = c(1, 2),
                      labels = c("Female", "Male"))
```

1. In Exercise 3 you learned how to pick out rows that meet a condition. You can use that skill to check whether your data are believable. This matters more than it sounds: a value that is simply wrong will not stop your code running, will not produce a warning, and will quietly change every result you calculate from it. Run `summary()` on your dataframe again and look at the minimum and maximum of each numeric variable. Three variables contain values that are not merely unusual but impossible. Two are impossibly low and one is impossibly high (hint: a body mass index below about 15 or above about 60 is essentially unheard of in a living adult, and nobody can survive with no cholesterol in their blood at all, so a blood fat concentration of exactly 0 is not a low reading, it is an impossible one). Find them, work out which patient the impossibly high one belongs to, and then set all of the offending values to `NA` so they cannot contaminate anything you calculate later (you will need the square bracket `[ ]` notation from Section 3.4.2, this time on the left hand side of an assignment). Re-run `summary()` afterwards to check it worked. Why is `NA` the right answer here, rather than deleting the whole patient record or guessing what the value should have been?

2. Another useful way to manipulate your dataframes is to sort the rows based on the value of a variable, or on a combination of variables. Rather counter-intuitively you should use the `order()` function to sort your dataframes, not the `sort()` function (see Section 3.4.3 of the Introduction to R book for an explanation). Ordering dataframes uses the same logic you practised in Q12 in Exercise 2. Sort all rows in the `cardiac` dataframe by ascending order of systolic pressure within each level of smoking status, and assign the result to a variable with a sensible name. The trick is to remember that `order()` will take more than one variable,

and that the order you give them in matters. Now take a look at the bottom of your sorted dataframe. Where have the patients with a missing smoking status ended up, and why?

3. Often, we would like to summarise variables by, for example, calculating a mean, median or counting the number of observations. To do this for a single variable it's fairly straight forward :

```
mean(cardiac$age)           # mean age
median(cardiac$systolic)    # median systolic blood pressure
length(cardiac$tchol)      # number of observations
```

3. (cont) Perhaps more interestingly, you might want to summarise one variable conditional on the level of another categorical variable, and to do several variables at once. The `aggregate()` function does exactly this (see Section 3.5 of the Introduction to R book, or `?aggregate`). Use `aggregate()` to calculate the mean age, systolic pressure, diastolic pressure and total cholesterol for each smoking category. Then calculate the same means for each combination of smoking category and sex.

The grouping variables go inside the `by` argument wrapped in `list()`, because a list is how `aggregate()` accepts more than one of them at once. Name the elements as you go, as in `list(smoking = cardiac$smoking)`, because those names become the column headings in the result and unnamed ones come out as `Group.1`.

A word of warning before you start. Remember those missing values you have been tripping over since Exercise 3? They have not gone away, and `tchol` still has two of them. Run your first `aggregate()` without any special handling, look hard at the `tchol` column, and only then reach for the `na.rm` argument.

4. Knowing how many observations are present for each category (or combinations of categories) is useful to determine whether you have an adequate sample size (for subsequent modelling for example). Use the `table()` function to determine the number of patients in each smoking category (see Section 3.5 again for more information). Next use the same function to display the number of patients for each combination of smoking category and sex. Does `table()` tell you about the patients whose smoking status is missing?

5. Your variables are measured on wildly different scales. Systolic pressure averages about 143 with a standard deviation of about 22, while total cholesterol averages about 7 with a standard deviation of about 1.3. That makes a question like ‘is this patient more unusual in their blood pressure or in their body mass index?’ impossible to answer from the raw numbers, because a difference of 10 means something completely different in each variable. Standardising fixes this. Subtract the mean from every value and divide by the standard deviation, and the variable is re-expressed as the number of standard deviations each patient sits from the average, whatever units it started in. These are called z-scores, and you will meet them again whenever you need to put variables on a common footing.

- a) Create two new variables in the `cardiac` dataframe, `bmi_z` and `systolic_z`, holding the standardised versions of `bmi` and `systolic`. You will need `mean()` and `sd()`, and by now you should know what to do about the missing values. Adding a column to a dataframe is covered in Section 3.4.4. Check your work by confirming that a standardised variable has a mean of 0 and a standard deviation of 1.
- b) Which patient sits furthest from the average body mass index, and how far out are they? (hint: `abs()` gives you the size of a number regardless of its sign, and `which.max()` tells you the position of the largest value). Look at that patient's `systolic_z` as well. Are they equally unusual on both measures?

- c) Now the important part. You set an impossible `bmi` of 514.6 to `NA` back in Q1. What would this standardisation have looked like if you had skipped that step?

Data linkage

Real analyses almost never involve a single file. The measurements you want are usually spread across several sources and you have to bring them together first. When those sources hold records about the same people, combining them into one dataset is called data linkage, and it is one of the defining techniques of health data science. Hospital admissions linked to prescribing records, a birth cohort linked to school attainment, a patient list linked to the death register. Each linkage answers a question that neither source could answer on its own, and this is the term we will use for it throughout the course.

The idea is simple enough. You match records on something that identifies the same person in both sources, usually an identifier such as a patient number. The details are where analyses quietly go wrong, which is what the next three questions are about. In R the function that does the work is `merge()` (see Section 3.4.5 of the Introduction to R book), and the operation it performs is called a join.

6. The patients in this study were followed up ten years after their first examination, and those follow-up measurements are held in a separate file. Download '*cardiac_followup.txt*' from the **Data** link, save it to your `data` directory and import it with `read.table()` into a variable called `followup`. How many rows does it have, and how does that compare with `cardiac`? Now use the `merge()` function to link the two together on the patient number, and assign the result to `cardiac_fu`. How many rows does the linked dataframe have, and can you explain why?

7. Losing 55 patients without being told is exactly the sort of thing that ruins an analysis, and it is why you check row counts every single time you link two sources. Often what you actually want is to keep every patient from the first dataframe whether or not they have a match in the second. Look at Section 3.4.5 and the help file for `merge()`, and find the argument that does this. Redo the linkage keeping all 163 patients, assign it to `cardiac_all`, and check how many of them have a missing follow-up systolic pressure. Finally, try running `merge(cardiac, followup)` without the `by` argument at all. It works. Why is relying on that a bad habit?

8. Ok, we have spent quite a bit of time (and energy) learning how to clean, summarise and link dataframes. The last thing we need to cover is how to export a dataframe from R to an external file (see Section 3.6 of the book for more details). Export your cleaned and linked dataframe `cardiac_all` to a file called '*cardiac_clean.txt*' in the `output` directory you created in Exercise 1. To do this you will need to use the `write.table()` function. You want to include the variable names in the first row of the file, but you don't want to include the row names. Also, make sure the file is a tab delimited file. Once you have created your file, try to open it in Microsoft Excel (or open source equivalent). Finally, and this is the part people forget: add a comment block at the top of your R script listing every change you made to the data and why. Which values did you set to `NA`, in which variables, and which patients did they belong to? Someone reading your work in six months, including you, needs to be able to answer that without re-running anything.

Optional extension: linking a lookup table

Only start this if you have finished everything above and have time to spare. It is not assessed, and nothing in the later exercises depends on it, but it is the most realistic linkage of the three and worth doing if you can.

There is one more kind of linkage worth knowing, and it solves the **smoking** problem that has been hanging over you since Exercise 3. Rather than typing the labels out by hand as you did for **Fsex**, you can keep the codes and their meanings in their own small file and link them on. Download '*smoking_lookup.txt*' from the **Data** link, save it to your **data** directory and import it into a variable called **smoke_codes**. Take a look at it. Notice that the column holding the code is called **code**, not **smoking**, so the two dataframes have no column name in common and **by** on its own will not work.

Look at Section 3.4.5 and the help file for **merge()** again to find the arguments you need (hint: **by** has a pair of siblings that let you name a different key in each dataframe. **merge()** calls its two dataframes **x** and **y**, in the order you supply them, so look for the two arguments whose names end in **.x** and **.y**). Link the labels onto **cardiac_all**, keeping all patients, re-export it as you did in Q8, and then check with **table()** that every patient has ended up with the right label.

End of Exercise 4